

CSE 2232: User Interface Development Lab

Credits	1
Contact Hours	28
Year	Second
Semester	Second
Prerequisite	CSE 1222: Object Oriented Programming Lab
Course Type	☐ Theory ☒ Laboratory work ☐ Project work ☐ Viva Voce

Motivation

Modern software is experienced largely through its user interface. This course introduces front-end development through HTML, CSS, JavaScript, UI/UX design, and React.js, enabling students to build accessible, responsive, and data-driven web interfaces.

Course Objective

This laboratory course provides hands-on experience in building modern web interfaces. Students will learn HTML, CSS, UI/UX principles, JavaScript, a CSS framework, basic HTTP/REST concepts, and React.js for consuming public APIs.

Course Outcomes (COs), Program Outcomes (POs) and Assessment

CO No.	CO Statement	Corresponding PO	Domain / Level of Learning Taxonomy	Delivery Methods & Activities	Assessment Tools
CO1	To reproduce wireframe designs as accessible, well-structured web pages using HTML, CSS, and UI/UX principles	Modern Tool Usage (PO5)	Psychomotor domain – Level 3	☐ Lecture Note ☒ Text Book ☐ Audio/Video ☒ Web Material ☒ Lab Manual	☒ CA ☒ Final Exam ☐ Assignment ☐ Notebook ☐ Presentation
CO2	To construct interactive and responsive UI components using JavaScript and a CSS framework	Modern Tool Usage (PO5)	Psychomotor domain – Level 4	☐ Lecture Note ☒ Text Book ☐ Audio/Video ☒ Web Material ☒ Lab Manual	☒ CA ☒ Final Exam ☐ Assignment ☐ Notebook ☐ Presentation
CO3	To develop a React.js web application that consumes public APIs using the Fetch API	Design/Development of Solutions (PO3)	Cognitive domain – Level 6	☐ Lecture Note ☒ Text Book ☐ Audio/Video ☒ Web Material ☒ Lab Manual	☒ CA ☒ Final Exam ☒ Assignment ☐ Notebook ☒ Presentation

Assessment and Marks Distribution

- Continuous Assessments (CA) (30%)
- A comprehensive final exam + Lab note book (60%)
- A class participation mark (10%).

Lab Course Contents / List of Experiments

- Lab 1 — HTML Fundamentals
 - Lab 2 — UI/UX Design Principles
 - Lab 3 — CSS Fundamentals
 - Lab 4 — Responsive Design with Bootstrap / Tailwind CSS
 - Lab 5 — JavaScript Basics, DOM and Form Validation
 - Lab 6 — Asynchronous JavaScript, HTTP/REST, and the Fetch API
 - Lab 7 — React.js Basics: Components, Props, State, and Hooks
 - Lab 8 — Public API Integration: Consuming REST APIs with React
 - Lab 9 — React.js Capstone Project
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Textbooks

1. Jon Duckett — *HTML and CSS: Design and Build Websites*, Wiley, 2011 (visual reference, still widely used)
2. Marijn Haverbeke — *Eloquent JavaScript* (4th ed.), No Starch Press, 2024 (free online at eloquentjavascript.net)